



Sachin Balu Khaire

Roll Number : 254159007
M.Tech - Biomedical Sciences and Engineering
Indian Institute of Technology Guwahati

+91-9022261911
sachinkhaire1451@gmail.com
s.khaire@iitg.ac.in
www.linkedin.com/in/skhaire2812

Education

Year	Degree / Certificate	Institute / Board	CGPA/Percentage
2025 - Present	M. Tech	Indian Institute of Technology, Guwahati	8.20 (Current)
2021-2025	B. Pharm	K. V. N. Naik Institute of Pharmacy, Nashik	8.00
2021	Higher Secondary	Maharashtra State Board	87.83%

Experience

- Industrial Trainee | Koral Pharma Pvt. Ltd.** *June 2024 - July 2024*
Assisted in pharmaceutical manufacturing processes and observed quality control procedures during industrial training.

Projects

- Smart Pressure Monitoring System for the Prevention of Pressure Ulcers** *Ongoing*
 - Designed a pressure monitoring system to detect prolonged pressure in bedridden patients.
 - Implemented a real-time alert system for detecting prolonged pressure exposure.
 - Aimed at early detection and prevention of pressure ulcers.
- Wearable Device for Early Prediction of Cardiovascular Risk using ML** *Ongoing*
 - Developed a wearable prototype for real-time cardiovascular monitoring.
 - Applied machine learning for early prediction of cardiac risk.
- Polyherbal Antipsoriatic Gel: Formulation and Evaluation** *Nov 2024 - May 2025*
 - Formulated a polyherbal topical gel using natural extracts for the management of psoriasis.
 - Evaluated the formulation for physicochemical properties, stability, and skin compatibility.

Technical skills

- Programming Languages :** Python
- Libraries:** NumPy, Pandas, Matplotlib
- Tools :** MS Excel, PowerPoint, MATLAB
- Instruments :** UV-Vis Spectroscopy, AFM, SEM, TEM, Raman Spectroscopy, XRD, HPLC

Positions of Responsibility

- Core Cultural Team Member:** K.V.N.Naik Institute of Pharmacy, Nashik *Nov 2024 - July 2025*

Achievements

- GPAT 2025 Qualified | Score: 195 | Percentile: 94
- GATE XL 2025 Qualified | Score: 503

Key Courses Taken

- Regulatory Affairs of Medical devices.
- Novel Drug Delivery System.
- Biotechniques and Bioinstrumentation Lab.
- BioMEMS and Biosensors.
- Biomedical Electronics and Instrumentation
- Biomaterials and Implants.